

Old UNM Algebra Qualifier Problems

Based on Chapter 4 of Dummit–Foote

Fifteen-year collection: January 2012–August 2023

Scope. This packet uses the third edition of Dummit and Foote, *Abstract Algebra*. Chapter 4 is “Group Actions”. The first section contains problems that can reasonably be attempted after this chapter. The second section records closely related problems.

I. Core problems from Chapter 4

January 2012 Qualifier, Problem 1

How many elements of order 7 must be in a simple group of order 168?

Solution. Let n_7 be the number of Sylow 7-subgroups. Since

$$168 = 2^3 \cdot 3 \cdot 7,$$

Sylow’s theorems give

$$n_7 \mid 24 \quad \text{and} \quad n_7 \equiv 1 \pmod{7}.$$

The only possibilities are $n_7 = 1$ and $n_7 = 8$. If $n_7 = 1$, the unique Sylow 7-subgroup is normal, contradicting simplicity. Hence $n_7 = 8$.

Every Sylow 7-subgroup is cyclic of order 7 and therefore contains six elements of order 7. Distinct subgroups of order 7 intersect only in the identity. Consequently the required number of elements is

$$8(7 - 1) = \boxed{48}.$$

January 2013 Qualifier, Problem 1

Let p be a prime. Show that every group with p^2 elements is abelian.

Note: We will see later that there are nonabelian group with p^3 elements.

Solution. Let G have order p^2 . The class equation for a finite p -group implies that its center is nontrivial, so $|Z(G)|$ is p or p^2 . If $|Z(G)| = p^2$, then $Z(G) = G$ and G is abelian. If $|Z(G)| = p$, then $G/Z(G)$ has order p and is cyclic. A group whose quotient by its center is cyclic is abelian: solved in [Homework 4](#).

January 2017 Qualifier, Problem 1

Suppose p and q are distinct primes such that

$$p \nmid q \pm 1 \quad \text{and} \quad q \nmid p \pm 1,$$

and let G be a group of order p^2q^2 . Show that G is abelian.

Solution. Let n_p and n_q denote the numbers of Sylow p - and q -subgroups. Sylow's theorems give

$$n_p \mid q^2, \quad n_p \equiv 1 \pmod{p}, \quad n_q \mid p^2, \quad n_q \equiv 1 \pmod{q}.$$

If $n_p \neq 1$, then $n_p = q$ or q^2 . The first possibility implies $p \mid q - 1$; the second implies

$$p \mid q^2 - 1 = (q - 1)(q + 1).$$

Since p is prime, either conclusion contradicts $p \nmid q \pm 1$. Therefore $n_p = 1$. The same argument, with p and q interchanged, gives $n_q = 1$.

Let P and Q be the unique Sylow subgroups. Both are normal, and $P \cap Q = \{e\}$. Moreover $[P, Q] \leq P \cap Q$, so P and Q centralize one another. Since $|PQ| = p^2q^2 = |G|$, we have

$$G = P \times Q.$$

Every group of order the square of a prime is abelian by the preceding problem. Thus both P and Q are abelian, and consequently so is G .

January 2017 Qualifier, Problem 2

Determine the number of Sylow p -subgroups in S_p .

Solution. The exponent of p in $p!$ is one, so a Sylow p -subgroup of S_p has order p . Such a subgroup is generated by a p -cycle. There are

$$\frac{p!}{p} = (p-1)!$$

p -cycles in S_p : list the p symbols cyclically and divide by the p cyclic rotations of a list. Each subgroup of order p has exactly $p-1$ nonidentity elements, and all of them are p -cycles. Conversely, a nonidentity p -cycle generates a unique subgroup of order p . Hence

$$n_p = \frac{(p-1)!}{p-1} = \boxed{(p-2)!}.$$

August 2017 Qualifier, Problem 3

Suppose p and q are distinct primes. Show that if

$$p \nmid q-1 \quad \text{and} \quad q \nmid p \pm 1,$$

then every group of order p^2q is abelian.

Solution. Let n_p and n_q be the Sylow numbers. Since

$$n_p \mid q, \quad n_p \equiv 1 \pmod{p},$$

we have $n_p = 1$ or q . The second possibility would imply $p \mid q-1$, contrary to the hypothesis. Thus the Sylow p -subgroup P is normal.

Similarly,

$$n_q \mid p^2, \quad n_q \equiv 1 \pmod{q}.$$

If $n_q \neq 1$, then $n_q = p$ or p^2 . The first possibility gives $q \mid p-1$; the second gives $q \mid p^2-1 = (p-1)(p+1)$. In either case $q \mid p-1$ or $q \mid p+1$, contradicting $q \nmid p \pm 1$. Hence the Sylow q -subgroup Q is also normal.

The normal subgroups P and Q have coprime orders, so they intersect trivially and centralize one another. Therefore $G = P \times Q$. Now P has order p^2 and is abelian, while Q has prime order q and is cyclic. Hence G is abelian.

January 2018 Qualifier, Problem 8

Let p be a prime, let G be a finite group whose order is divisible by p , and let P be a normal p -subgroup of G . Show that P is contained in every Sylow p -subgroup of G .

Solution. Let S be any Sylow p -subgroup of G . Because $P \trianglelefteq G$, the product PS is a subgroup of G . Its order is

$$|PS| = \frac{|P||S|}{|P \cap S|},$$

which is a power of p . Thus PS is a p -subgroup containing the Sylow p -subgroup S . By maximality of S among p -subgroups, $PS = S$. Consequently $P \leq S$. Since S was arbitrary, P is contained in every Sylow p -subgroup.

August 2019 Qualifier, Problem 5

Let p, q, r be primes with $p < q < r$. Show that a group G of order pqr has a normal Sylow subgroup of order p , q , or r .

Solution. Suppose, for contradiction, that none of the Sylow subgroups is normal. For the largest prime r , Sylow's theorems give

$$n_r \mid pq, \quad n_r \equiv 1 \pmod{r}, \quad n_r > 1.$$

The proper divisors p and q of pq are both smaller than r , so neither can be congruent to 1 modulo r . Hence $n_r = pq$. Distinct Sylow r -subgroups, each of order r , meet only in the identity. They therefore contribute

$$pq(r-1)$$

distinct nonidentity elements. Only

$$pqr - 1 - pq(r-1) = pq - 1$$

nonidentity elements remain.

Now $n_q > 1$, $n_q \mid pr$, and $n_q \equiv 1 \pmod{q}$. Since $p < q$, we cannot have $n_q = p$. It follows that $n_q \geq r$. Distinct Sylow q -subgroups contribute $n_q(q-1)$ distinct nonidentity elements, so they contribute at least $r(q-1)$ such elements. But $p \leq q-1$ and $r > q$, whence

$$r(q-1) > q(q-1) \geq pq > pq-1.$$

This exceeds the number of elements still available, a contradiction. Thus at least one Sylow subgroup is normal.

II. Closely related later-material problem

January 2018 Qualifier, Problem 9

Let p and q be distinct primes and let G be a group of order pq^r for some $r \geq 1$. If

$$q \nmid p-1 \quad \text{and} \quad p \nmid q^i-1 \quad \text{for every } 1 \leq i \leq r,$$

show that G is nilpotent. For what values of r must G be abelian?

January 2020 Qualifier, Problem 3

What is the order of a Sylow p -subgroup of S_{3p} for $p > 3$? Are the Sylow p -subgroups of S_{3p} abelian? Justify your answers.

August 2020 Qualifier, Problem 2

Consider $G = S_{28}$.

- (a) What is the order of a Sylow 7-subgroup of G ?
- (b) Is a Sylow 7-subgroup abelian? Justify your answer.

Source

University of New Mexico, Department of Mathematics and Statistics, [Past Qualifying Exams – Algebra](#). Dates and problem numbers are recorded above.